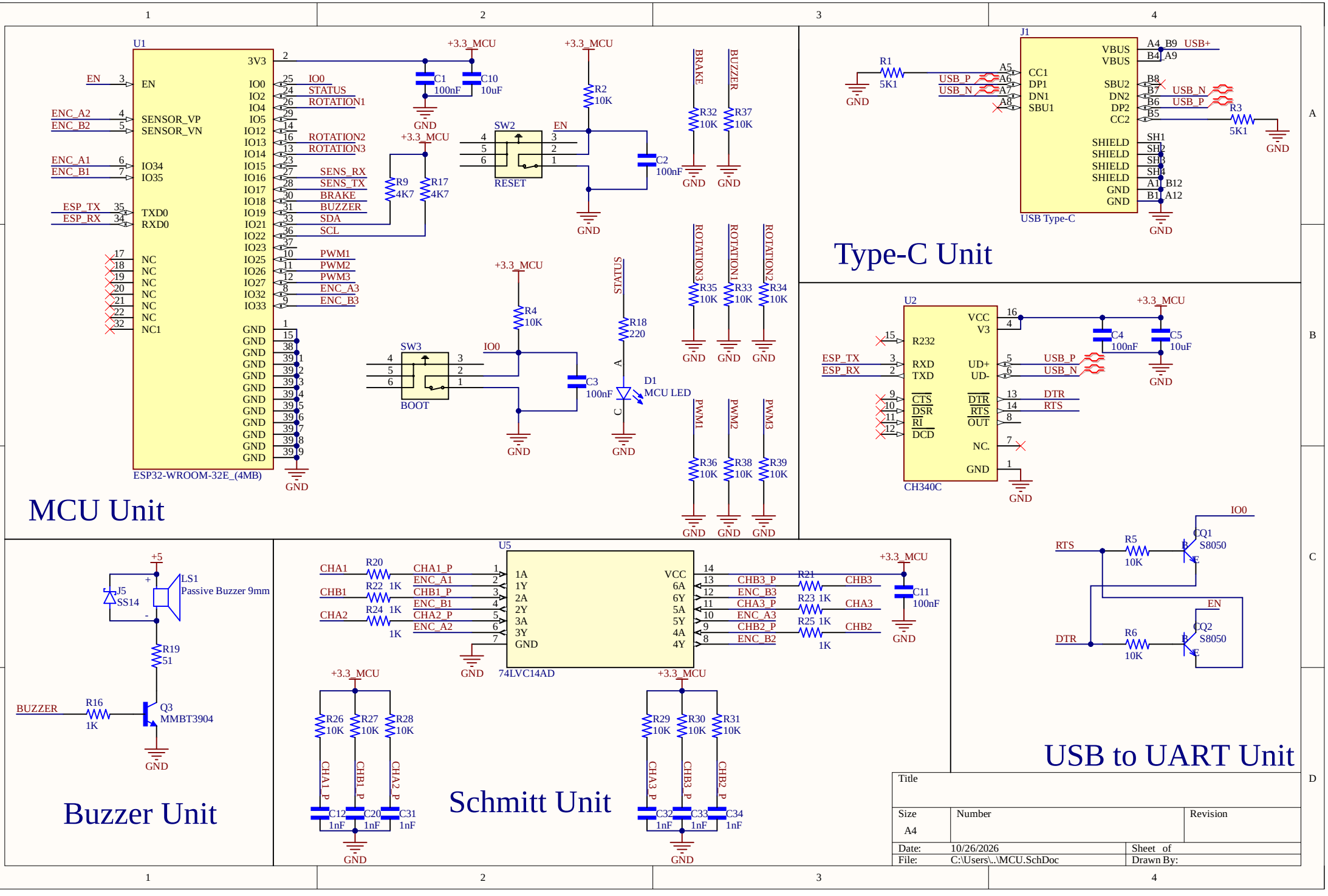
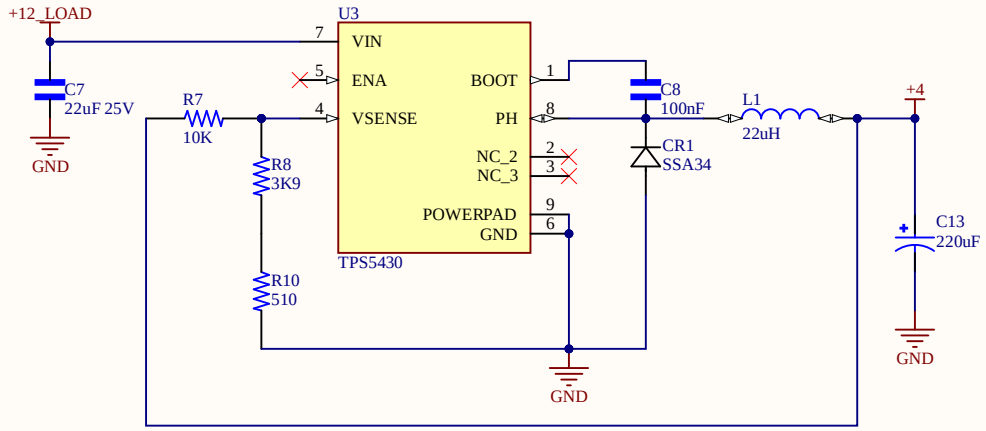
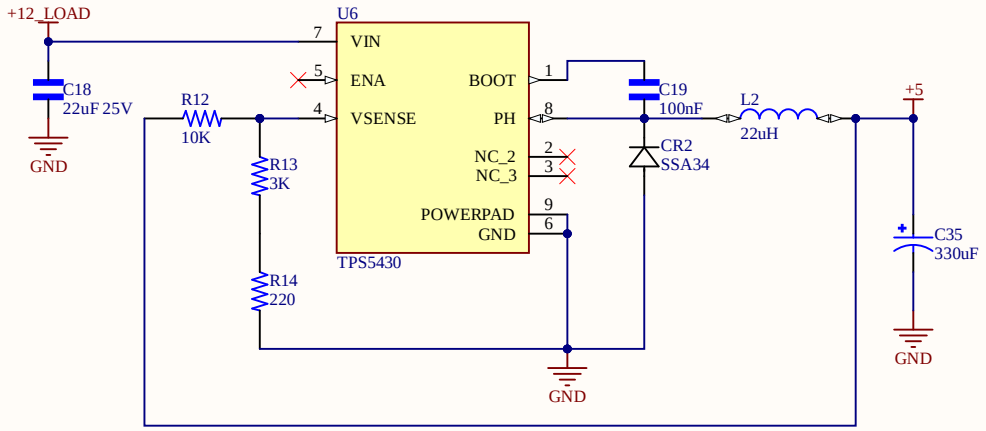
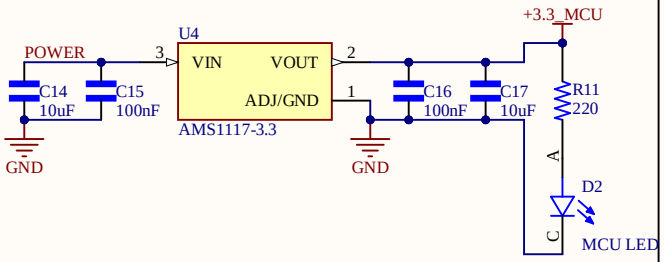

MCU Unit
Type-C Unit
USB to UART Unit
Buzzer Unit
Schmitt Unit
ESP32-WROOM-32E_(4MB)
CH340C
74LVC14AD
USB Type-C
Title
Size
A4
Number
Revision
Date: 10/26/2026
File: C:\Users\..\MCU.SchDoc
Sheet of
Drawn By:



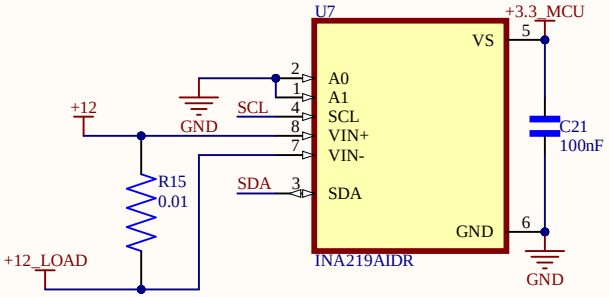
Buck DC-DC Control



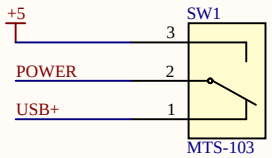
Buck DC-DC Motor Encoder



5 Voltage - LDO Unit

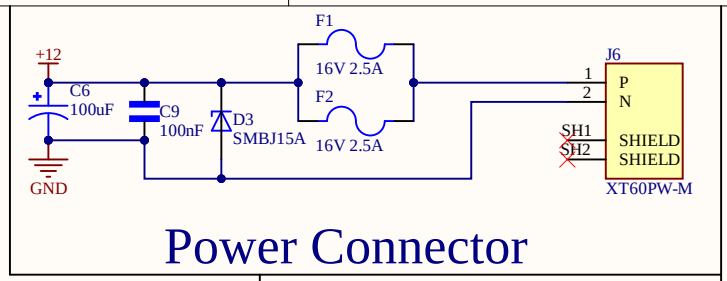


Battery Management Unit

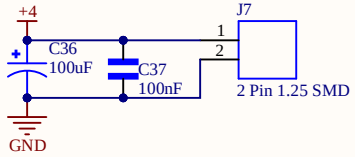


Switch Power Unit

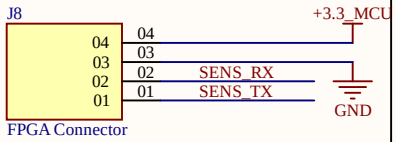
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Size: A4	Number	Revision
Date: 10/26/2026		Sheet of
File: C:\Users\..\Power.SchDoc		Drawn By:



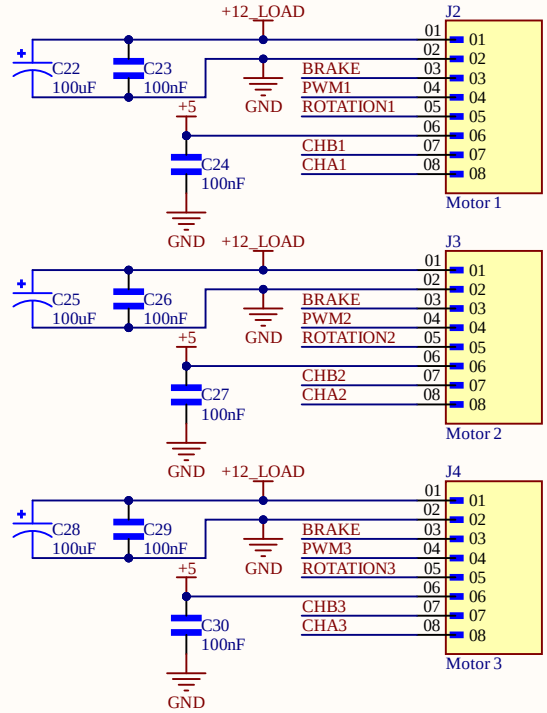
Power Connector



Power Sens Connector

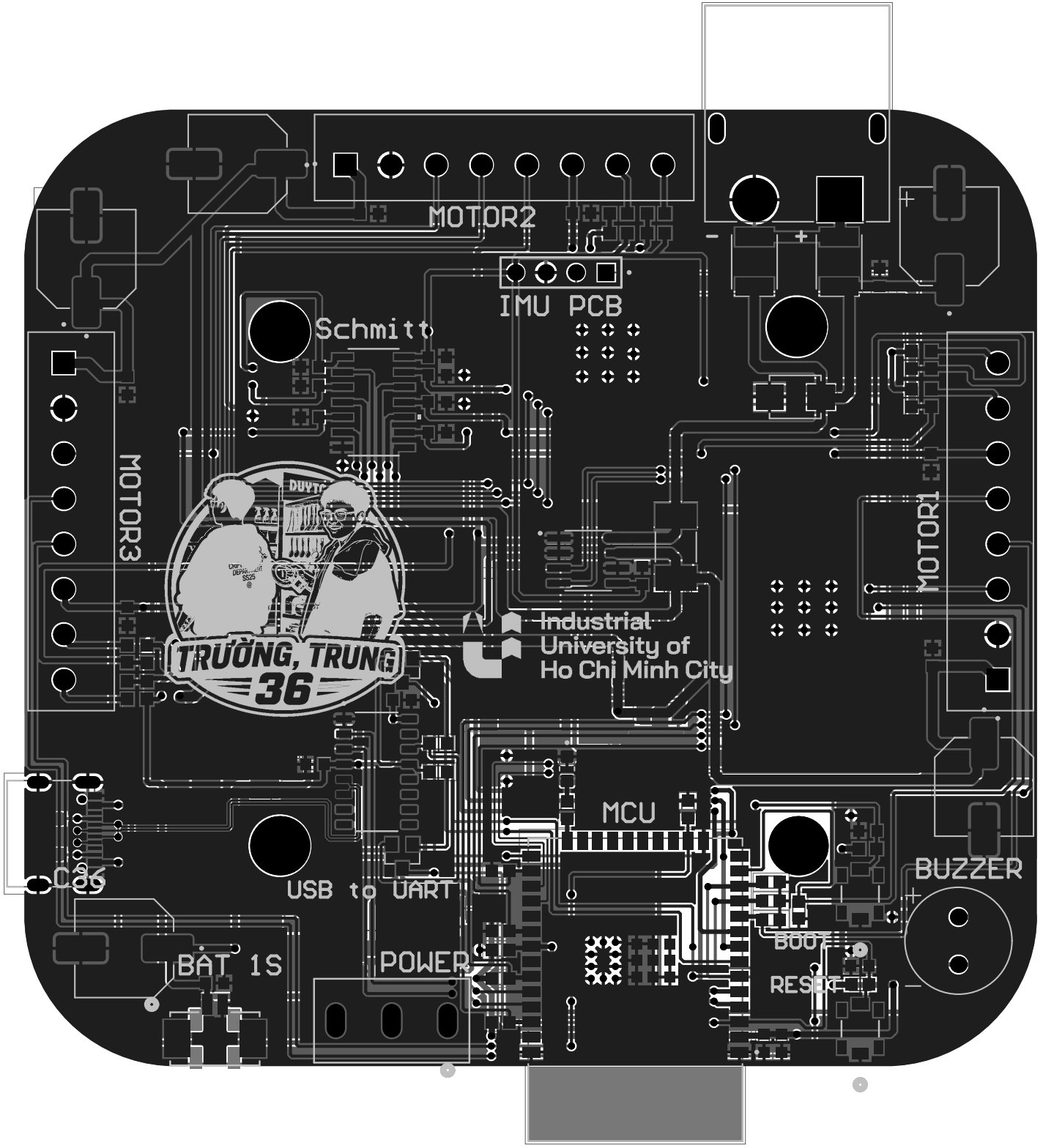


Sens Connector



Motor Connector Unit

Title		
Size: A4	Number	Revision
Date: 10/26/2026		Sheet of
File: C:\Users\..\Connector.SchDoc		Drawn By:


MOTOR2
IMU PCB
Schmitt
MOTOR3
MOTOR1
TRƯỜNG, TRUNG
36
Industrial
University of
Ho Chi Minh City
MCU
USB to UART
BUZZER
BAT 1S
POWER
BOOT
RESET

Comment	Description	Designator	Footprint	LibRef	Quantity
100nF	SMD Capacitor X7R(EIA) with Capacitance: 0.10uF Tol. 10%. Rated Voltage: 50Vdc.	C1, C2, C3, C4, C8, C9, C11, C15, C16, C19, C21, C23, C24, C26, C27, C29, C30, C37	CAPC1608X90N	GRM188R71H104KA9 3D	18
10uF	SMD Capacitor X7R(EIA) with Capacitance: 0.10uF Tol. 10%. Rated Voltage: 50Vdc.	C5, C10	CAPC1608X90N	GRM188R71H104KA9 3D	2
100uF	Capacitor Aluminum Electrolytic, 1000h 6.3 (Dia.) x 7.7mm, 100uF, 35V, WT Series	Nichicon UWT1V101MCL1GS	C6, C22, C25, C28, C36	CAPAE830X1000N	UWT1V101MCL1GS	5
22uF 25V		C7, C18	CAPC3216X229N	1206ZD226KAT2A	2
1nF	SMD Capacitor X7R(EIA) with Capacitance: 1000pF Tol. 10%. Rated Voltage: 50Vdc.	C12, C20, C31, C32, C33, C34	CAPC1608X90N	GRM188R71H102KA0 1D	6
220uF	220 ÂµF Molded Tantalum Capacitors 6.3 V 2312 (6032 Metric) 1.2Ohm @ 100kHz	C13	CAPPM6032X280N	T493C227K006CH641 0	1
10uF	10 ÂµF Molded Tantalum Capacitors 10 V 1206 (3216 Metric) 3.8Ohm	C14, C17	CAPC3216X110N	T491A106K010AT7280	2
330uF	100ÂµF 35V Aluminum Electrolytic Capacitors Radial, Can - SMD 80mOhm @ 100kHz 2000 Hrs @ 105Â°C	C35	CAP_EEEFP1V101AP	EEEFP1V101AP	1
SSA34	3A,40V,SMA SKY RECT	CR1, CR2	CR_A34-E3SAT_VIS	SSA34-E3SAT	2
MCU LED	Red 633nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D1, D2	LEDSC125X200X90-2N	S990110007F	2
SMBJ15A	600W, 17.6V, 5%, Unidirectional, TVS	D3	DIOM4336X265N	SMBJ15A	1
16V 2.5A	Polymeric PTC Resettable Fuse 16V 2.5 A Ih Surface Mount 1812 (4532 Metric), Concave	F1, F2	FUSER_MF-MSMF250/16X-2	MF-MSMF250/16X-2	2
USB Type-C		J1	AMPHENOL_10155435-00011LF	10155435-00011LF	1
Motor 1		J2	PHOENIX_1803484	1803484	1
Motor 2		J3	PHOENIX_1803484	1803484	1
Motor 3		J4	PHOENIX_1803484	1803484	1
SS14	Diode Schottky 40 V 1A Surface Mount DO-214AC (SMA)	J5	SMA	SS14	1
XT60PW-M	Socket, DC supply, male, PIN: 2	J6	AMASS_XT60PW-M	XT60PW-M	1
2 Pin 1.25 SMD	Connector Header Surface Mount, Right Angle 2 position 0.049 (1.25mm)	J7	CONN_532610001-3D_02_MOL	532610271	1
FPGA Connector		J8	SAMTEC_TSW-104-07-T-S-LL	TSW-104-07-T-S-LL	1
22uH	220 ÂµH Shielded - Inductor 1.45 A 338mOhm Max Nonstandard	L1, L2	INDPM12012X800N	CDRH127_LDNP-221MC	2
Passive Buzzer 9mm	AUDIO TRANSDUCER 3VP-P 70MA 20OHM 86DBA	LS1	XDCR_AT-0927-TT-2-R	AT-0927-TT-2-R	1
S8050	Bipolar (BJT) Transistor NPN 25 V 500 mA 150MHz 300 mW Surface Mount SOT-23	Q1, Q2	SOT-23	SMD-TRANSISTORS-NPN-25V-500MW(SOT-23)	2
MMBT3904	BJT, SOT-23, 40V, 200mA, NPN, 0.35W, 150Â°C	Q3	TRANS_MMBT3904	MMBT3904	1
5K1	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R1, R3	RESC1607X60N	RC0603FR-071K2L	2
10K	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R2, R4, R5, R6, R7, R12, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39	RESC1607X60N	RC0603FR-071K2L	20
3K9	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R8	RESC1607X60N	RC0603FR-071K2L	1
4K7	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R9, R17	RESC1607X60N	RC0603FR-071K2L	2
510	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R10	RESC1607X60N	RC0603FR-071K2L	1
220	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R11, R14, R18	RESC1607X60N	RC0603FR-071K2L	3
3K	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R13	RESC1607X60N	RC0603FR-071K2L	1
0.01	1 mOhms Â±1% 2W Chip Resistor 2512 (6432 Metric) Current Sense, Moisture Resistant Metal Element	R15	RESC6331X85N	CSNL2512FT1L00	1
1K	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R16, R20, R21, R22, R23, R24, R25	RESC1607X60N	RC0603FR-071K2L	7
51	1.2 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R19	RESC1607X60N	RC0603FR-071K2L	1
MTS-103	Description	SW1	MTS-103_JCV	MTS-103	1
RESET	Description	SW2	PTS820 J25K SMTR LFS_CNK	PTS820 J25K SMTR LFS	1
BOOT	Description	SW3	PTS820 J25K SMTR LFS_CNK	PTS820 J25K SMTR LFS	1
ESP32-WROOM-32E_(4MB)	Bluetooth, WiFi 802.11b/g/n, Bluetooth v4.2 +EDR, Class 1, 2 and 3 Transceiver Module 2.4GHz ~ 2.5GHz Integrated, Trace Surface Mount	U1	XCVR_ESP32-WROOM-32E_(4MB)	ESP32-WROOM-32E_(4MB)	1
CH340C	USB to serial chip CH340	U2	SOIC127P800X180-16N	CH340C	1
TPS5430	5.5V to 36V Input, 3A, 500kHz Step-Down Converter	U3, U6	CONV_TPS5430DDAR	TPS5430DDAR	2
AMS1117-3.3	Linear Voltage Regulator IC Positive Fixed 1 Output 1A SOT-223-3L	U4	SOT229P700X180-4N	AMS1117-3.3	1
74LVC14AD	Description	U5	SOT108-1_NXP	74LVC14AD	1
INA219AIDR	26-V 12-bit I2C output digital power monitor	U7	SOIC127P599X175-8N	INA219AIDR	1